



A field guide to
Brothers AM

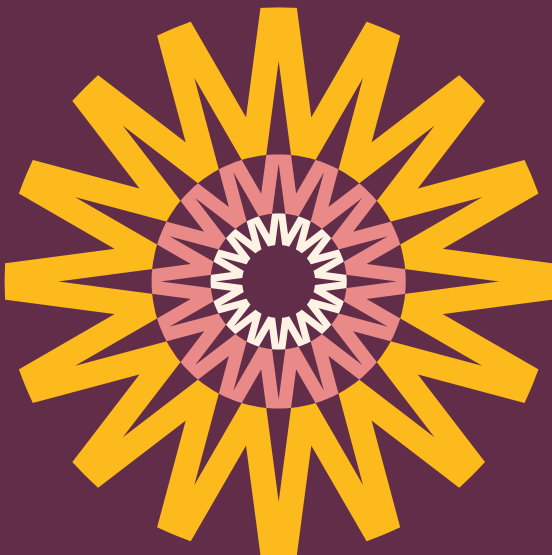


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9V DC Power req: 9V DC Center Negative ~200 mA

Foreword

From Joel Korte – Owner / Founder of Chase Bliss

For me, this story starts about five years ago. That's when I sent my first email to Mike Piera from Analog Man.

Our original Brothers pedal had been out for a few years by that time. I still love that pedal for many reasons, but while it can and does sound amazing, it can also be a challenge in some contexts to dial in a sound that inspires.

Then I got my hands on the King of Tone and it was the complete opposite. No matter what gear I was using, no matter what settings, it always sounded just right. It made me want to play guitar. I noticed how well it would map to the control scheme of Brothers, and from that moment onwards I was hooked. I became obsessed with the possibility of collaborating with Analog Mike.

Writing this now almost feels like waking up from a dream.

Did we really just make a pedal with Analog Man?

It's been an odyssey. We went all the way out to Tokyo to see firsthand where Mike would find vintage parts in the 80s when he was first starting Analog Man – along the way, rummaging through the same bins 40 years later, I found a part that would solve our final and biggest challenge on Brothers AM. It was truly unbelievable.

It's been that kind of project. I started this company over 12 years ago, and I genuinely can't imagine trying to explain this journey to that version of myself. I would not believe it was possible. I am so very grateful to Mike and his generosity, and the simple fact that Brothers AM exists.

Thank you,



From Mike Piera – Owner / Founder of Analog Man

It's hard to believe it's been five years. It took awhile to work out the bugs but we finally got here.

I've known about Joel and Chase Bliss for a long time and always felt they had a stellar reputation for honesty and service, which is unfortunately rare these days. I could feel that personal touch while working together. I'm not usually a fan of pedals with too many controls and settings, but this felt different. I've seen the way people navigate Chase Bliss pedals and the flexibility that all that control gives them.

It all became real for me when everyone finally visited about a year ago. I immediately felt at ease and on the same wavelength. We were invited to join Chase Bliss in Tokyo – which I'm sure Joel thought would never happen – but I was able to make some time and use a million frequent flyer miles to get there in good shape.

It was one of the best trips I've ever had, like a dream of my past. I hope this pedal will bring you the same warm feeling.

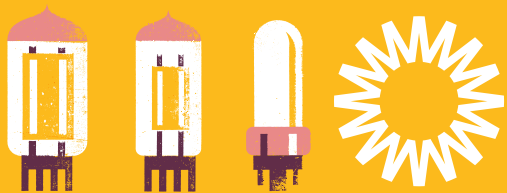
Enjoy!



chase bliss
ANALOG.MAN

Overview

Based on the legendary King of Tone overdrive and made with Analog Man himself, Brothers AM is a triple stack of saturation that always manages to sound just right.



Transparent overdrive is all about enhancing what's already there. It makes your instrument sing and your amp spark, but it also preserves their character. What makes the King of Tone so special – even 20 years later – is that it perfects the details: loud, clear, and responsive.

Brothers AM takes that coveted circuitry, gives you advanced control, and tosses in a couple exciting bonuses.

But it doesn't mess with what made it so good in the first place.

What it is.

Brothers AM is three circuits in series:

- Two identical, multi-mode gain channels
- A unique, two-voice treble booster



The stacking effects made possible by this chain of gain are extraordinary. Use it as intended to enjoy timeless transparent overdrive, or push it to create a crackling stream of overflowing gain.

It's all in there – prized and proven, and now able to go ever further.

Let's find out how.

WHAT'S NEW & DIFFERENT

- Master mode
- Presence link mode
- Bonus treble booster circuit
- Quick access to modes and advanced controls
- Expression, CV, and MIDI control
- Built-in presets

Setup

Here's some info to help you get Brothers AM settled into its new home.

POWER

Brothers AM requires a 9V DC, center negative power supply with at least 200 mA of current. Voltage higher than 9V risks damaging your pedal.

You'll see this symbol on your power supply:



OPTIONS

Brothers AM has a few ways to customize and fine-tune your experience. If you'd like to get right into all that, check out:

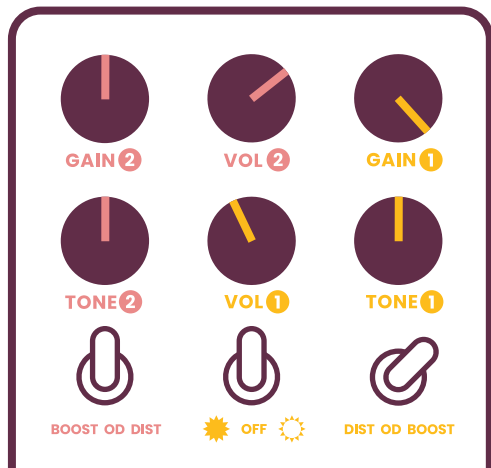
- Customize (pg. 28)
- External Control (pg. 30)

Otherwise it's probably best to start with all dip switches turned off.



Time to get warmed up.

Getting Started



Let's start here.



Turning on Channel 2 will give you a nice bit of crunchy, transparent overdrive. Every setup will be a little different, so feel free to make some adjustments and settle in.

You can think of Brothers AM as two gain pedals in series.

The first channel comes before the second and flows into it. This means you can use the first channel to push the second. Let's do that.



Channel 1 is acting as a boost to overload Channel 2, a handy way to enhance the energy but maintain the texture.



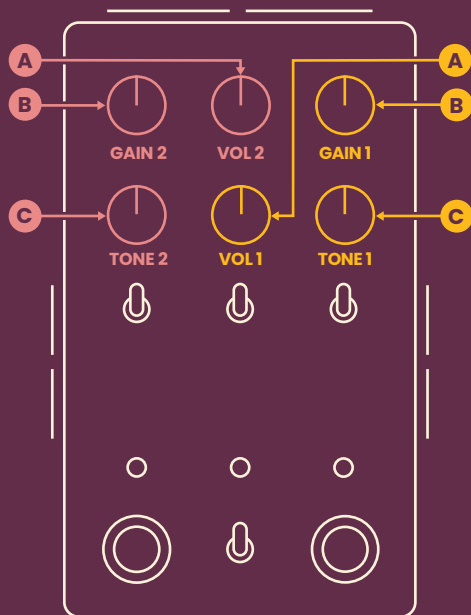
Let's go further and turn on the treble booster.

At this point you can cut through any mix.

This is one possible way to approach Brothers AM, using each of its circuits to level up bit by bit, but there are plenty more.

Let's dive in.

Controls – Knobs



Boost, drive, shape.

A

VOLUME

Controls the output loudness of the two independent channels.



When both channels are on, **VOL 2** sets the final output volume. You can now use the first channel to push the second further into saturation, but be mindful of the **VOL 1** setting – once Channel 2 is turned off it will jump to that volume. Brothers AM gives you the option of turning **VOL 2** into a master volume instead to help prevent volume spikes (pg. 29).

B

GAIN

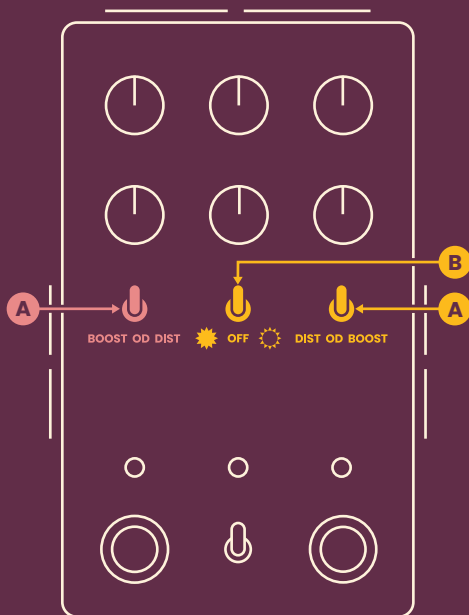
Controls the gain of each channel. This is responsible for the amount of saturation, but can also influence the overall level.

C

TONE

Controls the voicing of the channel, allowing you to brighten or darken the sound.

Controls – Modes



Select, engage, change.

A

MODE

Sets the style of circuit for each channel.

BOOST – A clean boost capable of low-gain overdrive. It has a clear sound and the loudest overall volume. Useful for characterful cleans, stacking with other boosts, or pushing an amplifier/overdrive.

OD/OVERDRIVE – The classic King of Tone sound – a soft-clipping overdrive with a natural, open character that preserves the tonality of your setup.

DIST/DISTORTION – A hard-clipping overdrive that has more compression and more saturation, but retains clarity. Useful for crafting more full-on distorted and blown out sounds.

You will notice the volume of each mode is different. This is the result of clipping – the more distorted the mode, the more compressed the sound. There is plenty of volume on tap to balance everything out, but be mindful of level jumps when switching between modes.

B

TREBLE BOOSTER

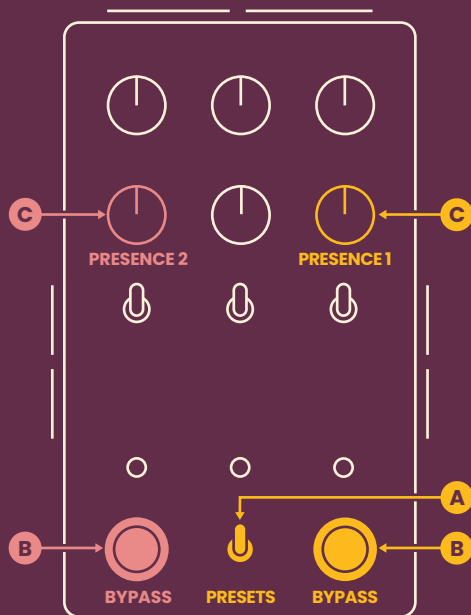
Engages a treble booster that pushes the upper mid and treble frequencies. The treble booster is linked to and engaged by Channel 1, and comes at the very front of the circuit.

 – A more cutting option, with increased brightness and treble.

OFF – No treble booster.

 – The classic “Rangemaster” sound, with heavy emphasis on the upper mids.

Controls – Everything Else



Open, stack, save.

A

PRESETS

The left and right positions each store a preset, while the middle position is live (current settings). See pg. 16 for more on presets, including how to save.

B

CHANNEL BYPASS

Tap to turn each of the channels on / off. Hold down to adjust PRESENCE (see below).



STANDARD



HI GAIN

The color of the LED will indicate if **HI GAIN** is turned on for that channel (pg. 28).

C

PRESENCE

This will let you boost the high frequencies for a more airy and brilliant sound. Each channel has a hidden PRESENCE control. To adjust it, hold the footswitch for that channel and adjust its TONE knob.

The recommended and default setting is turning **PRESENCE** all the way down.

Presets

Brothers AM lets you store and recall internal presets. All controls and knob positions – including the dip switches – are saved.

Presets are recalled using the toggle between the footswitches.



The left and right positions each contain a preset, and the middle position is a “live” state that matches the current controls of the pedal.



To save a preset to the right slot, hold down the right footswitch for three seconds, then add the left footswitch for another three seconds. The middle LED will blink to indicate success. Do the same process in the opposite order for the left slot.

Brothers AM also lets you access two additional presets by flipping on the BANK dip switch.

DEFAULT BANK

THE ANALOG MAN

HI GAIN 1

Mike Piera's preferred settings, adapted from his King of Tone. A solid starter for experiencing the pedal as intended. Kick on Channel 2 when you need a lift.

SUNNY SKIES

A mixture of everything Brothers AM has to offer. Treble boost into distortion into overdrive for a sound that is bright, focused, and charged.

ALT BANK

2-IN-1

A useful performance combo. Channel 2 is your always-on overdrive and Channel 1 is there to boost it beyond its limits. Turn on Channel 1 when you want more bite.

BAD BROS

HI GAIN 1, HI GAIN 2

As big and brash as Brothers AM can get. It's an overflowing distortion made by stacking and cranking everything.

Finding Your Sound

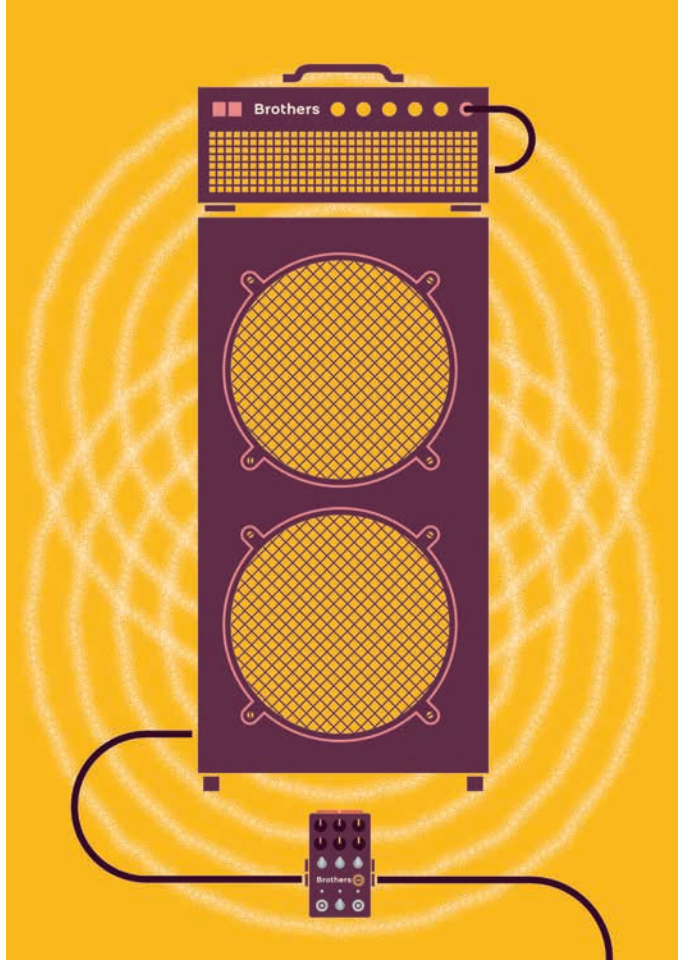
Brothers AM is a carefree pedal that sounds good in just about every position.

But it can do a whole bunch and you have choices.

So we're going to use the following pages to start exploring, show you what's possible, and give you some starter ideas.

**We'll start simple,
find a core overdrive sound,
and then work in the treble booster.**

Let's find the sound that you're looking for.



Two-Channel Ideas

The most straightforward way to use Brothers AM is as a two-channel effect. For example: One channel handles your edge-of-breakup sounds for the verse and the other has more blown-out sounds for the chorus.

Simply tap both footswitches at once and it will swap between the channels.



The benefit of this approach is that it's very predictable. Think of each channel like its own pedal and set it just as you like.

CLEAN CHANNEL, DIRTY CHANNEL



You can mimic a classic two-channel amp by using one channel for your clean sounds and the other for the distorted. Try using a low-gain overdrive on Channel 1 and a mid-gain distortion on Channel 2 – two clear, distinct stages, but with a common character.

AMP PUSHER



Use the **BOOST** mode on both channels with the same or similar settings, but set one to have a louder output. This “double boost” arrangement is useful if you prefer to make your amp's overdrive the focus.

You can take this idea further by engaging the treble booster on Channel 1, and/or pushing up the **GAIN**. This will give you one clean, transparent boost, and another that offers some edge and a richer character.

MOMENT MAKER



A two-channel approach is helpful if you want to create very specific moods within a set. Maybe you need an airy, nostalgic character for an intro, a sound that wouldn't work all the time but is just right to complete the moment. You can use one channel for that specialized sound and the other for your always-on sound.

Stacking Ideas

Things get a bit more fun once you start stacking.

In these examples, both channels are on at once and influence each other. The benefit of this approach is that you can find some massive and creative sounds as the two channels combine.

OVERLOADER

This is a good place to start because it's the most intuitive. Channel 2 is your primary always-on sound and Channel 1 is used to overload it.



A good starting point is selecting **OD** for Channel 2 and **BOOST** for Channel 1. Gradually turn up the volume on Channel 1 and notice how it then oversaturates Channel 2 and pushes it further than it can go on its own.

Leave Channel 1 off by default and pop it on when you need an extra charge.

EXPANDER

In this case we're using the first channel as our main voice and our second channel to lift and extend it.



Set the first channel to **OD** and the second channel to **BOOST** with the **TONE** knob turned up. In this case the second channel enhances the first, amplifying it and opening it up. Now you can turn on Channel 2 when you want to increase intensity without adding distortion.

COMBO

The final option is just to leave both channels on and combine them to create something new.



A versatile, loud, and open-sounding drive, created by setting both channels to **BOOST** and using Channel 1 to push Channel 2.

We recommend turning on the **MASTER** dip switch (pg. 29) while exploring boost stacking. This will let you turn up **VOL 1** without having disruptive jumps in volume.

Treble Booster

Brothers AM includes a bonus circuit:

A Rangemaster-style treble booster, inspired by the Beano Boost from Analog Man. This unique style of boost emphasizes the upper mid and treble frequencies to create a cutting sound ideal for lead tones.

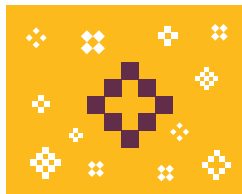


The original King of Tone did not have this Rangemaster-style circuit, so leave the toggle in the middle position if you want that classic sound.

Let's talk about how we can use it to go further.

Treble boosters are excellent tools for effortlessly standing out and creating big, biting moments.

Popular uses:



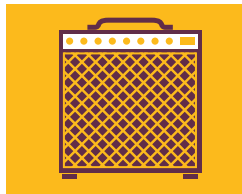
STAND OUT IN A MIX

Treble boosters emphasize just the right spot to make chords or leads stand out without the need to get more distorted.



BRIGHTEN UP

You can use a treble booster to lend a bit of jangle and sparkle to guitars with humbuckers or darker sounding instruments.



GET MORE FROM YOUR AMP

This was the original use of the treble booster. Bass frequencies overwhelm amplifiers and limit their range, so treble boosters were created to let players go further into overdrive by only pushing the treble and mids.



CREATE CUTTING LEAD TONES

Saving a treble boost for special moments is an effortless way to suddenly emerge from a mix and take the lead.

Treble Booster Settings

The following are a few ideas for getting started with the treble booster.



CUTTING CLEANS



This setting turns Channel 1 into something more like a traditional treble booster. Use this to push your amp or Channel 2, or to create a clean tone with a bit of edge that cuts through.



LIFTED DISTORTION



A brash, commanding distortion with heaps of harmonic overtones. It sounds even better with the **HI GAIN** dip switch turned on.



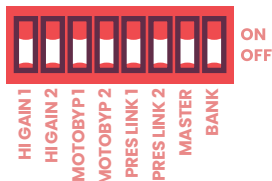
FULL STACK



This setting pours it all on. A big, snarling, clanging, electric tangle, pulled out of a transparent overdrive.



Customize



The **pink-labeled dip switches** on the top of your Brothers AM allow you to engage some alternate behaviors and set things to your liking.

HI GAIN

Increases the gain of all modes by about 25% (the circuit and character are otherwise identical). The bypass LED will now be red to indicate that **HI GAIN** is turned on.

MOTOBYP

“Momentary to bypass.” **MOTOBYP** lets you turn the **BYPASS** footswitch for either channel into a momentary control, meaning the pedal will only turn on when the footswitch is held down. For example, this might be useful for brief instances of boost.

You can also use it to momentarily bypass the pedal if you prefer – in this case simply turn on the **MOTOBYP** dip while the pedal is already engaged.

PRES LINK

Presence link. When engaged, the **TONE** knob will also simultaneously control the **PRESENCE** setting. This can be useful if you want an even more open and transparent sound, or when using full frequency instruments like a synth.

MASTER

Turns the **VOL 2** knob into a master volume control. **VOL 2** will now set the overall output volume at all times, even when Channel 2 is bypassed. This lets you explore different stacking combinations of the two channels while minimizing volume jumps. **VOL 1** will have no effect in this mode unless both channels are on. (More on stacking on pg. 22).

BANK

Switches over to an alternate bank of internal presets, giving you access to two additional slots on the pedal (Presets 3 and 4). Save and recall are performed as usual (pg. 16).

The following options can be turned on independently for either channel:

● **HI GAIN** ● **MOTOBYP** ● **PRES LINK**

The dip switch settings are saved with your presets.

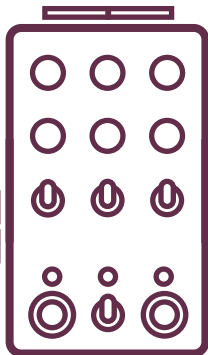
External Control

Brothers AM is compatible with MIDI, CV, expression, and external footswitch.

→ MIDI/EXT

MIDI: TRS cable

EXT: TS cable and normally open switch



CV range = 0–5V (higher voltage or any negative voltage could damage the pedal)

← EXP/CV

CV: Floating ring TRS to TS cable

EXP: TRS cable

CV and expression can be used to control the knobs on Brothers AM.

MIDI lets you go deeper and control everything, including the dip switches and presence settings.

CV and expression are both set up in the same way, using the dip switches on top of the pedal. You can control any combination of knobs and select the range and direction of the movement.

CV / EXPRESSION SETUP



1. Choose which knob(s) you wish to control.



2. Select the polarity. This will control the direction of the response, letting you either (R)ise or (F)all from the maximum point.



3. Set the range. The knob's position divides the range in two, and the SWEEP dip lets you select the (T)op or (B)ottom half.

MIDI requires a Chase Bliss MIDIbox to convert the signal to a ¼" TRS jack. For details on getting MIDI going with Brothers AM, please check out the MIDI manual.

The MIDI jack can also be used to bypass Channel 2 with an external pedal. Plug in any normally-open momentary footswitch using a TS cable and it takes control automatically. You're all set!

Internal Trimmers

There are three “trim pots” (also called trimmers) on Brothers AM’s circuit board that let you adjust the treble booster to taste. These adjustments are NOT required. We believe the default settings are just right for most people.



You need to be careful any time you change the position of a trimmer. You should use a very small flat head screwdriver and turn the trimmers gently. They are very easy to damage.

1 INPUT IMPEDANCE FOR TREBLE BOOSTER

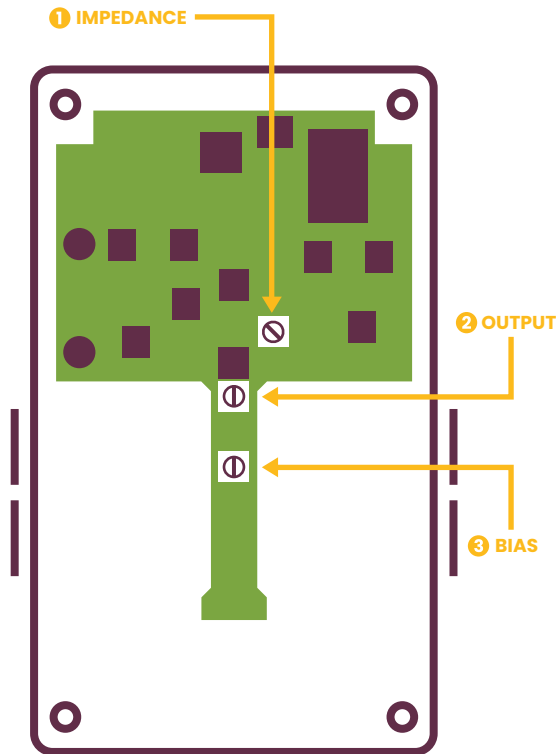
The treble booster included in Brothers AM is a Rangemaster-style circuit with a low input impedance. It’s happiest when connected directly to a guitar (no buffers). This trimmer lets you introduce some input impedance so that the circuit behaves much better after a buffer. The default setting is max CW (no additional impedance). To introduce some impedance, simply move this trimmer counter-clockwise. A good starting point after a typical buffer is noon (straight up and down), but feel free to experiment.

2 OUTPUT FOR TREBLE BOOSTER

This sets the level of the treble booster. Default is noon, and this is the setting that we prefer. It gives the normal KOT circuits a nice little bump. If you want the treble booster to hit those circuits harder, rotate the trimmer clockwise. If you want less and more of a “unity” type of feel, move it counter-clockwise.

3 BIAS FOR TREBLE BOOSTER

The default setting here is noon. This biases the transistor in a way that is consistent with the typical break-up characteristics of Rangemasters. You can squeeze more headroom out of this circuit and give it more of a symmetrical break-up by experimenting with the position of this trimmer. There are no specific rules here, try out some different settings and see if you have a preference.



(Default trimmer settings)



Goodbye.

**There's nothing more to learn.
Get out there and drive.**

Write us here any time
if you have questions:

help@chasebliss.com

We're happy to help.

Have fun!